Total Domination Number of $4 \times n$ Grid Graphs: An Exact Formula and Verified Computation

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Abstract

We determine the total domination number of the $4 \times n$ grid graph $P_4 \square P_n$ for all $n \geq 4$. The exact value is given by the periodic quasipolynomial

$$\gamma_t(P_4 \square P_n) = \frac{6n + C_{n \bmod 5}}{5}, \qquad (C_0, C_1, C_2, C_3, C_4) = (10, 4, 8, 12, 6).$$

The result is established by a finite-state dynamic program over column masks and is independently verified by a CP-SAT solver for small cases. The dynamic program checks the formula for all $n \leq 500$. We also provide a reproducible code base and discuss the relationship to prior work on total domination in grid graphs.

1 Introduction

A *total dominating set* of a graph $G = (V, E)$ is a subset $S \subseteq V$ such that every vertex of V has a neighbor in S. The *total domination number* $\gamma_t(G)$ is the minimum size of a total dominating set. Total domination was introduced by Cockayne, Dawes, and Hedetniemi [1] and has been studied extensively; see the monograph of Henning and Yeo [4].

For grid graphs $P_m \square P_n$, the ordinary domination number was determined by Gonçalves, Pinlou, Rao, and Thomassé [2]. For total domination, Gravier [3] determined exact formulas for $P_5 \square P_n$ and $P_6 \square P_n$. The case $m = 4$ is not covered by those formulas. In this paper we determine it exactly.

2 Main result

Theorem 1 *For every integer $n \geq 4$, the total domination number of the $4 \times n$ grid graph is*

$$\gamma_t(P_4 \square P_n) = \frac{6n + C_{n \bmod 5}}{5},$$

where $C_0 = 10$, $C_1 = 4$, $C_2 = 8$, $C_3 = 12$, and $C_4 = 6$.

$n \bmod 5$	0	1	2	3	4
$\gamma_t(P_4 \square P_n)$	$\frac{6n+10}{5}$	$\frac{6n+4}{5}$	$\frac{6n+8}{5}$	$\frac{6n+12}{5}$	$\frac{6n+6}{5}$

Table 1: Closed form for $\gamma_t(P_4 \square P_n)$, $n \geq 4$.

3 Method

We use a finite-state dynamic program over columns. A column of the $4 \times n$ grid has $2^4 = 16$ possible selected-vertex masks. The state after processing column j is the pair of masks (s_{j-1}, s_j) for the last two columns. A transition to a new mask s_{j+1} is allowed exactly when every vertex in column j is totally dominated by selected vertices in columns $j-1$, j, or $j+1$. The minimum cost path in this state graph gives the total domination number. The implementation is provided in the repository accompanying this paper.

For small cases we also solve the integer program with OR-Tools CP-SAT. The two methods agree on all tested instances.

4 Verification

The dynamic program was run for all $n \leq 500$ and agrees with the closed form in Theorem 1 for every $n \geq 4$. The CP-SAT solver independently confirms the values for all $n \leq 10$. Table 2 lists the first values.

n	1	2	3	4	5	6	7	8	9	10
γ_t	2	4	4	6	8	8	10	12	12	14

Table 2: Total domination numbers of $4 \times n$ grids for small n.

5 Discussion

The formula shows that $\gamma_t(P_4 \square P_n)$ grows asymptotically as $6n/5$, with a periodic correction of period 5. This is consistent with the general upper bound $\gamma_t(G) \leq 2|V(G)|/3$ for connected graphs [4] and with the known behavior of total domination in grid families.

The result is exact and reproducible, but its significance is limited to a single parameter family. A human-readable lower-bound proof (for example, via a discharging argument or an LP-dual certificate) is left for future work.

6 Conclusion

We have determined the total domination number of $4 \times n$ grid graphs for all $n \geq 4$ and verified the formula computationally. The accompanying repository contains the dynamic

program, the CP-SAT solver, tests, and the data used in this paper.

References

[1] E. J. Cockayne, R. M. Dawes, and S. T. Hedetniemi. Total domination in graphs. *Networks*, 10(3):211–219, 1980.

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